Deal-based censuses undercount private equity ownership of autism therapy: a reproducible, site-level national estimate

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Preprint, October 2026. Dataset version 2026.07-beta, resolver version 0.1.0, methodology version 2026.10-verification-v1.

Abstract

Background. The first peer-reviewed national estimate of private equity (PE) ownership of autism therapy sites in the United States counts 574 sites in 42 states as of December 2024, built from 147 acquisitions identified in a proprietary deal database. Deal-based censuses see a company at the moment it was bought. They cannot see the centres it opened afterwards.

Methods. We built a reproducible, site-level census of institutionally owned applied behavior analysis (ABA) clinics from four public source families: the federal NPPES bulk registry, operators' own published location directories and rosters, and hand-verified ownership facts. Every source document is fetched and content-hashed. A clinic is a physical site, keyed on canonicalized street and ZIP within an owner, never on a provider identifier. Where an operator publishes a complete directory it overrides the registry in both directions.

Results. Restricted to what the federal registry can see, we find 580 PE-owned sites, six from the published estimate, by a method sharing none of its data. Adding operators' own directories, we find 1,621 PE-owned clinics in the same 42 states, 2.8 times the deal-based count. A second outside estimate, counted by platform rather than by deal, agrees with ours to within 8% over the 16 platforms both cover (1,521 against 1,642). PE holds 2.8% of the 21,088 ABA sites in the registry nationally, a floor, but 15.8% in Minnesota, 13.2% in New Mexico and 9.5% in Arizona. The registry's blind spot varies by operator from none at all to complete: one 18-centre operator is entirely invisible to it. A hand check of a seeded random sample found 66 of 67 PE clinics correct on site, address, brand and owner (98.5%, 95% CI 92.0% to 99.7%).

Conclusions. Censuses of PE ownership sort by unit rather than by rigor. Counts built from deals or registrations see a company frozen at acquisition; counts built from platforms or directories see it today, and the difference is a factor of nearly three. Neither a deal list nor the provider registry is a sufficient basis for oversight of a sector whose operators keep growing after they are bought.

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1. Background

Applied behavior analysis (ABA) therapy for autistic children has, over the past decade, become an active area of private equity investment: 117 of the 147 PE acquisitions of autism service providers identified by the first national study fell between 2018 and 2022 [5]. The pattern is the familiar roll-up: a sponsor buys a regional operator as a platform, adds smaller practices to it, and opens new centres under the platform brand. The largest operator in this dataset publishes 414 centres. The case most often cited when this model is questioned is the Center for Autism and Related Disorders, then the largest US autism therapy chain, which Blackstone acquired around 2018. It filed for Chapter 11 in 2023 having closed more than 100 locations [7], and the bankruptcy court approved its sale in pieces later that year [8].

The question has a public-finance dimension because much of this care is paid for by Medicaid. The Department of Health and Human Services Office of Inspector General (OIG) has audited state Medicaid ABA programs in a continuing series, and each published audit has found improper payments at scale: at least $77.8 million in Colorado (2022-2023) [1], $56 million in Indiana (2019-2020) [2], $18.5 million confirmed and a further $94.3 million potentially improper in Wisconsin (2021-2022) [3], and $45.6 million in Maine (2019-2023) [4]. Spending grew fast over the audited periods: Colorado's from $60.1 million in 2019 to $163.5 million in 2023, Maine's from $52.2 million to $80.6 million over the same years. None of these audits attributes a dollar to any company or type of owner. They audit state programs, and the federal provider registry that they and most other analyses rely on records the entity that bills, not the firm that owns it.

Two recent studies have measured PE ownership in this sector directly. Arnold and colleagues, in *JAMA Pediatrics* in January 2026 [5], identified 147 PE acquisitions of autism service providers between 2015 and 2024 in the PitchBook database, verified and expanded them from press releases and company websites, and counted 574 PE-owned service delivery sites across 42 states as of 31 December 2024. The authors name two limitations themselves: they cannot report PE's share of all autism service sites, and they are likely undercounting. In April 2026 the Private Equity Stakeholder Project (PESP) published a report on PE in ABA and Medicaid [6] whose appendix lists 34 PE-backed ABA platforms with a facility count for each, drawn from PitchBook and LevinPro. Its leading recommendation is greater ownership transparency and facility-level oversight.

Both studies rest on proprietary deal databases, and neither publishes its site list, so neither can be reproduced, updated or checked line by line. This paper offers a census built entirely from public, fetchable sources, at the level of the physical site, with every source document content-hashed and every number regenerable from a published archive. It addresses both limitations the JAMA authors name: Section 4.3 reports PE's share of all ABA sites in the registry, and Sections 4.5 and 4.6 measure the undercount and locate where it comes from.

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2. Data

Fundprint draws on four source families. Every record in every family is fetched over HTTP, stored verbatim as a content-addressed blob, and linked to the claims it supports before any claim is eligible for publication. A row whose source document cannot be retrieved and hashed does not enter the dataset, regardless of how plausible the underlying fact is. This rule has real cost: two ownership claims that are well documented in secondary reporting remain unpublished because their primary sources return HTTP 403 to every client attempted, and one state Medicaid audit is published with its reasoning and without its dollar figures for the same reason.

2.1 The federal provider registry

The backbone is the National Plan and Provider Enumeration System (NPPES), the CMS registry that assigns a National Provider Identifier (NPI) to every provider that bills federal health programs. Fundprint uses the **monthly bulk dissemination archive**, not the NPPES API, and the distinction is consequential rather than a matter of convenience.

The API caps a query at 1,200 records and returns only an NPI's *primary* practice address. A chain that registers one NPI and attaches thirty locations to it as secondary addresses is therefore visible to the API as one address. The bulk archive ships a separate Practice Location Reference File (`pl_pfile`) containing exactly those non-primary locations, which is where a multi-site operator's other centers actually live. Any census built from the API, or from a taxonomy query against it, is structurally blind to this file. So is any census built from company websites alone, for the opposite reason: a website lists what the company chooses to advertise today.

The archive used for this release is approximately 1.1 GB compressed, expanding to an 11.5 GB CSV. It is streamed out of the zip rather than extracted, and only the filtered ABA extract, a few megabytes, is retained as a snapshot alongside the upstream URL and the SHA-256 of the archive itself:

```
archive_sha256 = e73068bbafb9ed72bb9303b36a1091ddce66f6e510a99fa1ca3339a04954cfea
```

Publishing that hash is what makes the registry-based results in Section 4 reproducible by a reader with no access to anything proprietary. The archive is free, requires no login, and is not governed by a data use agreement.

A provider organization enters the ABA universe if it holds a live, non-deactivated organization NPI and either carries a behavior-analysis taxonomy or bears the name of a tracked ABA brand. The second clause is necessary because several genuine ABA chains register under no ABA taxonomy at all. Omitting them would drop tracked clinics out of the denominator while leaving them in the numerator, inflating the share.

2.2 Owner location directories

The second family is the operator's own published list of centers. Where a chain publishes a machine-readable directory, Fundprint reads each center's address from the page's structured data: a schema.org `MedicalBusiness` block, a WordPress REST collection, a store-locator endpoint, or the

site's semantic per-field address markup. This is a read of published structured data rather than brittle scraping of rendered prose.

This family is the single largest methodological difference between this work and a deal-based or registry-based census, and it accounts for the majority of the published dataset: **1,575 of 1,757 clinics are attested by an owner's own current directory or roster; 182 rest on the registry alone.**

A directory must be read at the leaf page. An earlier release recorded Autism Learning Partners as operating a single clinic because its state and county index pages carry no street addresses. The 59 pages beneath those indexes each publish one address. The operator is now recorded at 45 clinics. "This directory has no addresses" is a claim about the page that was actually opened, and the page to open is the last one.

Directory records are self-reported by the owner, which is a weakness for some questions and a strength for the one that matters most here: whether a center is open now. A registration is filed once and essentially never revoked. A directory is what the company tells prospective clients today.

2.3 Owner rosters

Some operators expose their entire center list through a single structured endpoint, typically a WordPress custom post type or the WP Store Locator plugin's `store_search` action with `autoload=1`. One request then returns every center with a clean address and, in some cases, the sub-brand. This is treated as a distinct source family because attribution is free: a roster published by the owner is the owner asserting the center is theirs, so no name matching is required and rows stage under the owner's own name.

2.4 Curated ownership facts

Ownership links themselves come from hand-verified primary sources: an acquisition announcement, a portfolio page, or reputable trade-press reporting that explicitly states *current* ownership. Each such fact fetches and content-hashes its own source URL. Historical ownership that has ended is recorded as a dated event with its own source and displayed on a timeline, never as a live ownership link.

Two acquisition tools are worth noting as negative results. A Bureau of Labor listing source and an SEC EDGAR acquirer were both built and are both dormant: the first because its content is rendered client-side, the second because it stages to a table no resolver reads. They are documented here so that a reader does not assume the pipeline draws on filings it does not draw on.

2.5 Sources considered and rejected

State Medicaid provider files were researched and rejected. Only one state publishes a usable bulk file, and its `robots.txt` disallows all automated access. More importantly, a Medicaid-based denominator would bias *against* the paper's own finding rather than for it: private-equity-backed chains skew commercial, and the largest operator in this dataset does not accept Medicaid in its home state at all.

Building fifty state scrapers would have produced a denominator that understates private-equity presence.

TLS-level access control shapes what is fetchable in a way worth recording. Several sources classify clients by TLS handshake fingerprint rather than by user agent, returning HTTP 403 to a Python HTTP client and HTTP 200 to `curl` presenting an identical user-agent string. The fetch layer retries such blocks through `curl` while continuing to present Fundprint's real user agent and contact address, and continuing to honor `robots.txt`. No browser is impersonated, and sources that block all clients are recorded as blocked rather than worked around.

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3. Method

3.1 The unit of analysis is a physical site

The central methodological commitment is that **a clinic is a physical service location**: not an NPI, not an office, not a brand. A site is identified by

```
site_key = (owner_entity, normalize_street(address_line1), zip5)
```

with a fallback to `(owner_entity, city, state)` for directory rows that publish no street.

Keying on the NPI instead would fail in both directions simultaneously, which is why no NPI-based de-duplication appears anywhere in the pipeline. Several NPIs can share one address: one operator holds six NPIs at a single Broomfield, Colorado suite under two legal-entity name variants, and an early pass that counted registrations reported 184 rows for 71 real sites, inflating that operator roughly threefold. And one NPI can hold many addresses, which is precisely how chains register secondary centers in `pl_pfile`. A count of registrations is therefore not merely noisy; it is biased unevenly across operators, which makes per-firm counts non-comparable.

3.2 Address canonicalization, and why it is not fuzzy matching

`normalize_street` canonicalizes USPS street suffixes, directionals and unit markers, so that `AVE` and `Avenue`, and `STE 101`, `Suite 101` and `#101`, all key alike. Before this existed, the site key stripped punctuation only, so the registry's `18301 N 79TH AVE, BUILDING A STE 101` and the owner directory's `18301 N 79th Avenue, Building A, Suite 101` were two clinics. **Forty-four real sites were double-counted this way** across six operators, once per source.

Two properties of this transform are load-bearing:

1. **It is a canonicalization, never a fuzzy match.** Every pair it folds together is an abbreviation and its own expansion. A looser rule, in particular one that dropped the unit designator, would merge two genuinely distinct clinics in one office park, which is the exact failure the site key exists to prevent.
2. **It applies to addresses only.** A separate and weaker `normalize` handles brand names, where collapsing "st" into "street" would be meaningless and risky. Using the address normalizer on

names, or the name normalizer on addresses, are both errors that have occurred and both produce wrong counts.

A related rule governs address *parsing*: **a wrong street is worse than no street**, because a wrong street produces a wrong site key, which stops the same building matching itself across two sources and counts it twice. An address with no comma before the city, such as `7041 Transit Rd East Amherst New York 14051`, cannot be split by rule, since peeling trailing words carries the street name into the city and leaves the street as a bare house number. Where a source names the locality separately, the parser subtracts the known locality rather than guessing. Where it does not, the parser refuses the row. The same rule applies to markup: a directory card that writes `421 Windchime Place<br> Colorado Springs, CO` marks the end of the street with the line break, and a parser that flattens the card to text before splitting it loses that boundary. One parser did exactly this, misfiling eight of one operator's 30 centres, and the garbled addresses then caused the directory rule of Section 3.4 to retire five correctly addressed registry rows. The hand verification of Section 5 caught it; the parser now splits on the markup and refuses a card it cannot split that way. Relatedly, a directory entry is not always an address at all: one operator lists a city with the text "In-home services available now" in the street position, so a street is required to carry a house number.

3.3 One suite is one clinic, across brands and across sources

A financial sponsor that acquires two chains and merges them leaves both registrations standing at the same address. Three such cases appear in this release, covering sites in Florida, Wisconsin and Pennsylvania, and **23 sites were double-counted** before they were collapsed. Where a collapse is required, the surviving row is chosen on evidence: an owner's own directory beats a registry record, and a fresher registration beats a staler one.

The reasoning here is worth stating explicitly, because an earlier version of this work disclosed the problem as a residual limitation rather than correcting it, on the theory that merging two separately registered legal entities was a stronger claim than the evidence supported. That is backwards. **Declining to merge is not the neutral option.** It asserts that two competing centers operate in one suite, which is the claim the evidence contradicts. When the choice is between two positive claims, "publish less" means the one that counts fewer clinics, not the one that requires less thought.

A second, subtler collapse operates within a single owner family. When one sponsor's five brands share one directory, the registry has often filed the same suites without a unit designator, so a naive merge of directory and registry rows mints a second clinic per suite spelling. Corroboration is therefore performed at *building* level while identity remains at *suite* level, and net-new buildings alone are staged. The correction is directory-count-aware: a building the directory lists once collapses to one clinic, while a building the directory lists twice, such as an address in Dublin, Ohio holding two of the family's brands in different suites, remains two. The rule can therefore never merge two real suites.

3.4 The owner's own directory overrides the registry, in both directions

Where an operator publishes a *complete* directory of its centers, and the registry disagrees about whether a center exists, the directory wins. This is the largest single lever in the dataset, and its effect is bidirectional.

The clearest case is the largest operator tracked. Its own directory lists 414 centers; the registry carried roughly half that many rows, and was wrong in both directions at once. The registry carried 206 live rows for the operator, and 105 of them are addresses the operator's directory does not list. It **missed most of Texas**, where the operator lists 227 centers and the registry carried 15, because a chain does not separately register every center it opens. And it **invented centers that do not exist**: nine in Ohio, three in Virginia and one in Georgia, all states the operator does not serve; an apartment; the operator's own corporate headquarters, filed as a practice location on the very NPI carrying its 109 real centers as secondary locations; and 34 stale Colorado registrations, where the registry gave 67 against the operator's own 42. After correction the operator publishes at **412 centers**, two short of its own directory, the difference being rows the address parser refused rather than guessed at.

Applied across ten operators, the rule corrects counts in both directions: 212 to 412, 145 to 101, 90 to 70, 1 to 45, 54 to 53, 24 to 12, 18 to 64, 37 to 74, 41 to 54, and 29 to 30. **Every one of them was wrong in both directions.**

The rule fixes three distinct faults with no special-casing for any of them: closed centers that were never deregistered, head offices filed as practice locations, and registrations belonging to entirely different companies whose names merely share a prefix with a tracked brand. On the third fault, a name-prefix match cannot distinguish a tracked chain from an unrelated counseling practice, a home-care company, or a chiropractor sharing the first two words of its name. Before this rule, such businesses were being published as private-equity-owned autism clinics.

Two constraints keep the rule honest. It runs **only where the directory is complete**, since a regional directory would quarantine real clinics elsewhere, and completeness is checked rather than assumed. And it **quarantines rather than deletes**: the registration is real, and what is false is the inference that a registration is a center. A center that genuinely exists survives via its directory row.

3.5 Scope rules

Three scope tests exclude records that are real but are not ABA therapy clinics.

In-home providers operate no clinics. An in-home operator still holds NPIs and must register them somewhere, so the registry returns an address that looks like a clinic and is not one: one operator registers exactly one office suite per state across fourteen states, and another registers at apartments. Such an operator is published with an ownership record, a clinic count of zero, and an explicit label. Deleting it would understate private-equity presence in ABA, so the ownership is retained even though the clinic count is not.

A multi-service provider is in scope only when ABA is a core line. The test is the registrations, not the company's self-description. One pediatric therapy provider carries a behavior-analysis taxonomy

at 22 of 24 sites and is included; one behavioral-health company has one primary-ABA identifier out of 65 and is excluded. The test cuts one way only: it can exclude a multi-service company, and it can never include a chain that would otherwise be out of scope.

Non-therapy sites are excluded. One operator's stand-alone diagnostic and evaluation clinics are held out, because this dataset counts therapy centers. Another operator's "coming soon" listings are excluded, because a leased address is not an operating clinic.

Head-office exclusions are made **only on direct evidence**, never on a heuristic, and only two appear in this release, each because the owner's own directory omits the address. A tempting rule, that many NPIs at one address indicates a head office, was tested against the data and found false: the largest tracked operator registers three to four NPIs at every one of its 35 Colorado centers. Applying that rule would have deleted dozens of real clinics. The negative result is recorded because the rule is intuitive enough to be reinvented.

3.6 The ownership firewall

The owner table is populated by scraping sponsors' portfolio pages, so it contains every company a tracked sponsor owns, including optometry, dental and restaurant chains. These were harmless against a taxonomy-filtered API query and are not harmless against the bulk registry, where they match: an early unguarded pass attributed roughly 5,000 optometry, dental and fast-food locations to private-equity firms as autism clinics.

An explicit `is_aba` flag, defaulting to false, is therefore the firewall: only owners marked as ABA operators can capture clinics. A second flag, `directory_only`, marks brands whose names are too generic to be safely name-matched against the registry at all; these are linkable from their own rosters and are never used to match registry records.

A related constraint governs naming. The registry carries an operator's **legal** name, which is frequently not its consumer brand: one operator registers under its search-fund holding company's name, another under a vocational-services entity, a third under a regional legal name it no longer trades as. The name-prefix linker must match what the registry carries, so `owner_entity.name` holds the legal name and a separate `trade_name` holds the brand. Published views display `COALESCE(trade_name, name)`. Renaming the legal field to the friendly name breaks the linker silently, and has done so.

3.7 Validation, floors and quarantine

Resolution proposes; validation disposes. Every claim carries a confidence score and a method tag recording how the link was established (`human_verified`, `exact_match`, `llm_inferred`, or `fuzzy_high`). A chain's confidence is the **minimum** along the chain, never the product or the average, so a weak link defines the chain.

Publication requires clearing a per-claim-type floor defined by the methodology version: 0.85 for clinic existence, 0.80 for clinic-to-owner, 0.85 for owner-to-parent, and 0.75 for a dated acquisition event. A claim below its floor is retained internally and waits for better evidence; it is never deleted.

Corrections are quarantines, never deletions. A correction opens a validation run and writes a `quarantined` decision per affected claim. The claim stays on the record and leaves the published views. Every validation run produces an append-only audit record: run identifier, timestamp, input claim identifiers, the resolver and methodology versions in force, and the deciding rule per claim. A reader asking "how did you know this at the time" must be answerable from that record without reconstruction.

It is worth recording that although a language-model resolution path exists in the codebase, **it is dormant: zero published claims in this release were established by it.** Everything published here is deterministic.

3.8 Denominator and share

The denominator is computed from the same registry archive as the numerator, in **one pass over one universe with one site key**, so the numerator is a strict subset of the denominator by construction.

An earlier implementation built the numerator by name-matching the archive directly rather than reading the published dataset. It therefore saw none of the corrections above: it counted centers that had just been quarantined as closed, and it counted the name-collision companies of Section 3.4. One state was published at 7.6% private-equity penetration on the strength of eight centers the named operator does not run. The numerator now reads the published clinic table and intersects it with the registry universe, so a single correction moves the map, the operator tables and the national share together.

Three further properties of this computation matter:

Distinct sites are counted as a union, never as a sum of per-operator sizes. An address carrying two brands or two legal entities would otherwise count twice on both sides, making the numerator a multiset rather than a subset. This is not a corner case: roughly 2,000 addresses in the registry are shared between legal entities. The visible consequence is that the operator-size distribution in Section 4, being per-operator, sums to 21,756 site-slots against 21,088 distinct sites, a difference of 668 shared addresses.

The visibility test operates at building level while identity remains at suite level. The registry frequently records `551 36th St SE Unit 2` where the owner writes `551 36th St. SE`. Requiring an exact site-key match to consider a published clinic registry-visible dropped 94 real sites, 82 of them private-equity-owned. The test therefore peels unit markers, and resolves to the *archive's* key rather than Fundprint's, so the numerator remains a strict subset. A building that is ambiguous in the archive, holding several suites, is skipped rather than guessed. This relaxation applies to the visibility test only. It does not relax `site_key`, under which two suites in one office park remain two clinics.

The published share is a floor and is labelled one. Closed centers are removed from the numerator, because the operators' own directories can be read, and cannot be removed from the denominator, because the other seventeen thousand operators' directories cannot. The bias runs against the finding. It is disclosed rather than adjusted away, since adjusting it would mean guessing.

3.9 What is deliberately not computed

An earlier release headlined private equity's share of clinics run by operators with five or more locations. It was withdrawn, and it is not reinstated here, for one disqualifying reason and two supporting ones.

The denominator was endogenous. An operator is a "chain" *because* it has many sites, and it has many sites *because* private equity rolled it up. Private equity's own buying inflated numerator and denominator together. A sponsor that acquired forty four-site operators and merged them into one forty-site chain would barely move its share of chain-run clinics while its actual market power grew enormously: it manufactures its own denominator. The measure was partly blind to the thing it existed to measure. Supporting the same conclusion, the five-site cutoff was arbitrary and undefended, and no other published work uses the measure, so it could not be checked against anything.

The general form of the trap is worth naming, because it is easy to re-derive under a new name: **any share of "the part of the market private equity could plausibly buy" has this same defect**, because private equity defines that part.

What replaces it requires no threshold and no editorial choice: an absolute count, a share of the full registry universe, per-state shares, and the operator-size distribution published in full so that any cut is the reader's and is stated out loud.

3.10 Confidence as four questions

A single confidence label conflates questions that fail independently, so each published clinic carries four graded dimensions rather than one: whether it is **open**, whether the **address** is right, whether it is a center rather than in-home care (**site type**), and who **owns** it. An overall grade is the weakest of the four. All four are derived from published fields alone, source types, registry freshness, service model and ownership-claim provenance, so a reader can reproduce the grade rather than trust it.

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4. Results

4.1 The count

The published dataset holds **1,757 ABA clinics in 42 states owned by institutional financial owners. Private equity owns 1,621 of them.** The other 136 are held by a pension fund (70), a family office (30) and two search-fund owners (36). They are published under their own labels and never folded into the PE figure, which is the like-for-like quantity for every comparison below.

A count needs no denominator, which is why it is the headline: nobody can hand it a wrong one. It is nonetheless a floor, because 11 known PE-backed platforms, holding 504 facilities by an outside count, have not yet been read at all (Section 6). The one error that runs the other way, a closed centre the registry still lists, is confined to the registry-only clinics and is graded below.

Of the 1,757 clinics, **1,575 are attested by the owner's own current directory or roster**; 182 rest on the federal registry alone. Each registry-only clinic is graded by the age of its registration record: 117 were re-certified within three years, 35 between three and six years ago, and 30 six or more years ago. That last group, 1.7% of the dataset, is where a closed centre is most likely to remain published.

4.2 Concentration among sponsors

Fifteen PE sponsors own the 1,621 PE clinics, and ownership is concentrated. One sponsor's single platform accounts for a quarter of the total, and the three largest sponsors hold 47%.

Sponsor	Clinics	Principal platform(s)
Charlesbank	412	Action Behavior Centers
KKR	197	BlueSprig family (four brands)
Gryphon Investors	151	LEARN Behavioral (federation of named brands)
Tenex Capital Management	139	Behavioral Innovations, Monarch
Nautic Partners	128	Proud Moments
Arsenal Capital Partners	101	Hopebridge
GTCR	79	Caravel Autism Health
General Atlantic	75	ACES
Petra Capital Partners	74	ABS Kids
Six further sponsors	265	

4.3 Share of the market, and why it is a floor

The registry archive lists **17,569 ABA organizations operating 21,088 distinct sites**. Restricted to sites the registry can see, so that numerator and denominator come from one source under one site key, Fundprint can name the owner of 686 (3.3%), and **580 (2.8%) are PE-owned**. Clinics known only from an owner's directory are excluded from both sides of this share, because the registry cannot see them; that is why the share's numerator is far smaller than the count.

The national figure is small because the profession is overwhelmingly independent:

Sites per operator	Operators	Sites
1	15,142	13,940
2 to 4	2,145	4,867
5 to 9	192	1,142
10 to 24	76	1,033
25 or more	14	774

Sites are counted distinctly within each class; across classes they sum to 21,756 against 21,088 distinct sites, because 668 addresses are shared between operators in different classes (Section 3.8). The dis-

tribution is published in full so that a reader who wants a "share of chains" draws the line themselves and says where; Section 3.9 explains why this paper does not.

4.4 Concentration is local

Care is bought locally, so the national share understates what a family sees. Among the 50 jurisdictions with at least 25 ABA sites in the registry, PE's share of sites is highest in:

State	PE sites	ABA sites	PE share
MN	22	139	15.8%
NM	14	106	13.2%
AZ	50	529	9.5%
CO	54	688	7.8%
PA	33	431	7.7%
WY	2	27	7.4%
ID	4	58	6.9%
MD	30	464	6.5%
GA	42	792	5.3%
IL	30	562	5.3%

These are registry-basis shares and so are floors on the same terms as the national figure. Wyoming and Idaho rest on small counts and are shown for completeness rather than emphasis.

4.5 Reconciliation with the deal-based estimate

The JAMA estimate counts 574 PE-owned sites as of December 2024; this dataset counts 1,621 as of October 2026, 2.8 times as many. Asserting that one figure is right would be unhelpful. The difference can instead be decomposed, and the order of the decomposition matters.

First, the footprint agrees. Both studies find PE-owned sites in exactly 42 states. The disagreement is about depth, not about where PE operates.

Second, restricted to the registry, the two counts converge. Of the 1,621 PE clinics, 580 are visible in the federal registry: six sites from the deal-based 574, by two methods sharing no data source. That convergence is what makes the remainder interpretable rather than a standoff between two unreproducible numbers.

Third, the remaining 1,041 clinics come from a source neither the deal list nor the registry consults: the operator's own directory. They are centres the operator says it runs today, at addresses it publishes, that no registration records.

Fourth, the registry's blind spot is not a constant. Measured operator by operator, the ratio of published centres to registry-visible centres runs from 1.0 (several operators register every centre) to 21.3 (one operator publishes 64 centres, of which the registry sees 3), and is unbounded for one:

an 18-centre Wisconsin operator has no registry presence at all. No single multiplier can correct a registry-based or deal-based count, because the correction differs by operator by more than an order of magnitude.

The JAMA letter reports its five largest states, which allows a partial comparison:

State	JAMA (Dec 2024)	Fundprint PE (Oct 2026)	Ratio
CA	97	148	1.5
TX	81	379	4.7
CO	38	67	1.8
IL	36	52	1.4
FL	36	80	2.2

The ratio is largest in Texas, where the largest operator in the dataset lists 227 centres of its own, of which the registry carried 15. Part of every ratio is real growth over the 19 months between the two counts, and that part cannot be separated without the JAMA study's site list, which is unpublished (Section 6).

4.6 A second estimate, in a different unit

One ratio against one estimate is the weakest form of this argument: it cannot distinguish a deal list that undercounts from a pipeline that inflates. The PESP appendix provides a test, because it counts in a third unit, facilities per platform, from a source that is current rather than frozen at the deal.

Over the 16 platforms both sources measure, **PESP counts 1,521 facilities and this dataset 1,642, a ratio of 1.08**. The differences run in both directions: this dataset is higher on 10 platforms, lower on 4, and exact on 2 (70 and 70; 64 and 64). The comparison includes three platforms whose owners are not PE (a pension fund, a family office and a search fund), which is why its total exceeds the PE-only 1,621. A seventeenth platform, an in-home provider PESP lists with 22 facilities, is excluded as a scope rule rather than a disagreement: this dataset counts no clinics for an operator that runs none.

The 1.08 is a precision check over shared platforms, not a national total, and must not be read as deflating the 2.8. Nationally the three figures are: the deal list, 574; this dataset, 1,621; and PESP's appendix summed over all 32 in-scope platforms, 2,047. **The two outside estimates differ from each other by more than either differs from this one, and this dataset sits between them, nearer the platform count.** It sits below PESP chiefly because it has not yet read the platforms that make up the difference (Section 6).

The independence of this check is partial and is stated as such. This dataset adopted PESP's appendix as its coverage denominator, so the list of platforms is shared; the counts are not, since PESP's come from commercial databases and these from directories and the registry. The units also differ at the margin: PESP counts centres "open or opening soon", this dataset counts operating sites.

The pattern across all four counts is the finding. Deal-based (574) and registry-based (580) counts both see a company as it was when it was bought or registered. Platform-based (1,521) and directory-based (1,642) counts both see it today. A chain bought with 40 centres that now runs 400 is counted at 40 by the first kind and 400 by the second. The censuses sort by unit, not by rigor, and the factor of nearly three between the kinds is a property of the unit rather than of anyone's diligence.

4.7 What the auditors can see

The registry's blind spot is not only a measurement problem. The registry is the standard public record of who provides care, and it is the view available to anyone without access to commercial deal databases. In the four states with published OIG audits, this dataset tracks 156 PE-owned clinics: 67 in Colorado (54 visible in the registry), 41 in Indiana (11 visible), 48 in Wisconsin (2 visible) and none in Maine. **In Wisconsin, the federal registry can see 2 of the 48 PE-owned centres in the state.**

This pairing supports no causal claim, and none is made. No audit attributes any payment to PE or to any owner, and Maine, with the third-largest finding of the four ($45.6 million) and no PE-owned clinics tracked, shows directly that these audits are not measuring ownership. What the pairing does show needs no causal claim: a large, fast-growing public expenditure is being audited in states where the public provider record does not show who owns the providers and, in Wisconsin, does not show most of the PE-owned centres at all. Facility-level ownership transparency, the leading recommendation of the advocacy report [6], is the gap this dataset is built to fill.

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5. Validation

A seeded, stratified random sample was drawn from the dataset as it stood on 19 July 2026 (seed 913063207): 110 PE clinics, 20 clinics held by other financial owners, and 20 registry ABA sites that the dataset attributes to nobody. Within the first two strata the draw was split by source in proportion, because registry and directory rows fail differently. The third stratum is the only one that can detect a false negative; a sample drawn only from published rows can measure precision but can never find a missed owner. Rows were shuffled before review, and review stopped at 99 of the 150, which are therefore themselves a random subset of the draw.

Each published clinic was checked against sources independent of the one that produced it, and counted as correct only if all four held: the centre is open, at the address given, under the brand given, and owned by the firm given. A clinic the reviewer could not confirm counts as a failure.

66 of 67 PE clinics were correct on all four questions: 98.5% (95% Wilson interval 92.0% to 99.7%). By source, 48 of 48 directory-sourced PE clinics were correct (92.6% to 100.0%) and 18 of 19 registry-sourced (94.7%, 75.4% to 99.1%). The single failure is a registry-only row at an address the operator's own directory does not list, which is where the design predicts failures to fall. Pooled with the non-PE strata the figure is 83 of 84 (98.8%), but the non-PE stratum was deliberately oversampled to test the owner-type label and so is not pooled into the headline.

No sampled clinic was closed or was not an ABA clinic (0 of 84; upper bound 4.4%). None of the 15 unattributed registry sites reviewed had a financial owner the dataset had missed. That is evidence of coverage rather than proof of it: at n = 15 the upper bound on the miss rate is 20.4%.

Two reviewer verdicts were changed after review. In each, the reviewer marked a clinic as being at the wrong address, and the operator's own directory showed it runs two centres in that city, of which the reviewer had found the other. Before those adjudications the PE rate was 64 of 67 (95.5%). Both figures are published, and each changed row carries its reason in the committed review file.

The review also caught an error that every automated check had passed: one operator's directory parser was misfiling street names into the city field (Section 3.2). It affected eight centres, was fixed, and the correction was published with the reason. This is the case for hand verification as a standing step rather than a one-off: the garbled rows were each internally consistent, and only a person reading the address against the world could see they were wrong.

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6. Limitations

The two counts are 19 months apart. The deal-based estimate is as of 31 December 2024 and this dataset as of October 2026. Part of the 2.8-fold difference is real growth in that interval, and it cannot be separated from the difference in method without the JAMA study's site list, which is not published. This paper does not claim the whole factor as method. The registry-basis convergence (580 against 574) and the platform-basis agreement (1,642 against 1,521, counted six months apart rather than nineteen) are the evidence that method, rather than growth alone, drives most of it.

The registry reports existence-ever, not existence-now. A provider registration is filed once and essentially never revoked; every published registry clinic reports active status, including some last certified many years ago, and chains do not deactivate an identifier when a centre closes. Owner directories remove this problem for 1,575 clinics. For the 182 registry-only clinics it is graded rather than assumed away, and 30 of them sit on records six or more years old. The verification sample found no closed clinics, but the registry-only strata are small.

The count is a floor, for a second and independent reason. This dataset covers 21 of the 32 in-scope PE-backed ABA platforms in PESP's appendix. The 11 it has not yet read, 8 not started and 3 blocked on sources that cannot be fetched, hold 504 facilities by PESP's count, including the post-bankruptcy Center for Autism and Related Disorders (105). Reading them moves this dataset toward the platform count, not away from the deal count.

The share is a floor, and the bias runs against the finding. Closed centres are removed from the numerator, because tracked operators' directories can be read, but cannot be removed from the denominator, because the other 17,000 operators' cannot. The asymmetry understates PE's share. It is disclosed rather than corrected, because correcting it would mean guessing.

Directories are self-reported. An operator's directory is the best available evidence of which centres are open now, which is why it overrides the registry, but it is the operator's own statement. Rules ex-

clude what a directory lists but does not operate: "coming soon" sites, diagnostic-only clinics, service-area pages and in-home offices. A directory overrides the registry only where it has been checked as complete.

Ownership is controlling ownership at a point in time. Minority stakes in which founders retain control are excluded, which removes several recognized investor relationships from the count. Ownership changes; each link cites a source stating current ownership, and a former owner appears as a dated event, never as a live link.

The verification has one reviewer. The sample was reviewed by the author alone, so no inter-rater agreement can be reported. The seed, the drawn sample and the labelled review file are published so the exercise can be replicated independently, which is invited. The sample also predates the current release, and rows corrected since the draw are not re-sampled.

What is deliberately not reported. An earlier release headlined PE's share of clinics run by operators with five or more sites. It was withdrawn because its denominator is endogenous to PE's own acquisitions (Section 3.9), and no equivalent measure is reported here under another name.

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7. Reproducibility

Every number in this paper is regenerated from public inputs by public code.

- **Code and data pipeline:** https://github.com/fundprint/fundprint-data
- **Methodology, versioned:** https://github.com/fundprint/fundprint-methodology (this paper corresponds to `2026.10-verification-v1`)
- **Public dataset and dashboard:** https://whofundsmytherapist.com, built from https://github.com/fundprint/fundprint-dashboard
- **Registry archive:** the CMS NPPES monthly dissemination file, whose SHA-256 is given in Section 2.1
- **Archived dataset:** Zenodo, https://doi.org/10.5281/zenodo.23426864

Each source document behind a published claim is stored as a content-addressed snapshot and cited by hash, so a reader can confirm that the page a claim rests on said what the claim says on the day it was read. The verification sample's seed, draw and labelled review file are in the data repository under `samples/` and `data/verification/`.

The full regeneration sequence, from archive to published figures, is:

```
python scripts/run_acquire_nppes_bulk.py --archive monthly.zip
python scripts/run_acquire_roster.py
python scripts/run_acquire_curated.py
python scripts/run_validate.py --resolver-version 0.1.0 \
    --methodology-version 2026.06-floors-v0
python scripts/create_dashboard_views.py
python scripts/compute_market_share.py --archive monthly.zip
```

```
python scripts/build_platform_denominator.py
python scripts/build_reconciliation.py
python scripts/build_state_files.py
python scripts/score_verification.py data/verification/reviewed_3bf05f0a.csv \
    --sample samples/verify_3bf05f0a.json --out data/verification/verification.json
python scripts/export_dashboard_snapshot.py
```

Declarations

Competing interests. The author declares no competing interests. The author holds no position in, and receives no compensation from, any company, sponsor or advocacy organization named in this paper.

Funding. This work received no specific funding.

Ethics. This study uses only publicly available records about organizations: a federal provider registry, companies' published location directories, public ownership announcements and published government audits. It involves no human participants and no individual-level or patient data, and so did not require ethics review.

Data availability. The full dataset, the reviewed verification sample and every derived figure are openly available under CC BY 4.0 at https://github.com/fundprint/fundprint-data and archived on Zenodo at https://doi.org/10.5281/zenodo.23426864. The registry archive is the free CMS NPPES monthly dissemination file, identified by SHA-256 in Section 7. Archived copies of third-party source documents are retained as evidence and are not redistributed under that license.

Code availability. All code is openly available under the MIT License at https://github.com/fundprint/fundprint-data and https://github.com/fundprint/fundprint-dashboard. The methodology is versioned and published under CC BY 4.0 at https://github.com/fundprint/fundprint-methodology.

Author contributions. A.D. conceived the study, built the data pipeline, performed the hand verification, analyzed the data and wrote the manuscript.

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